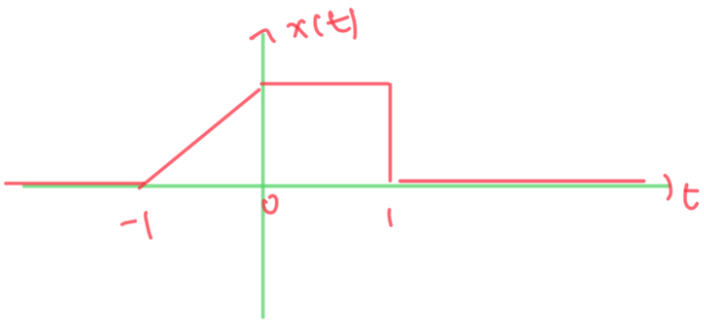
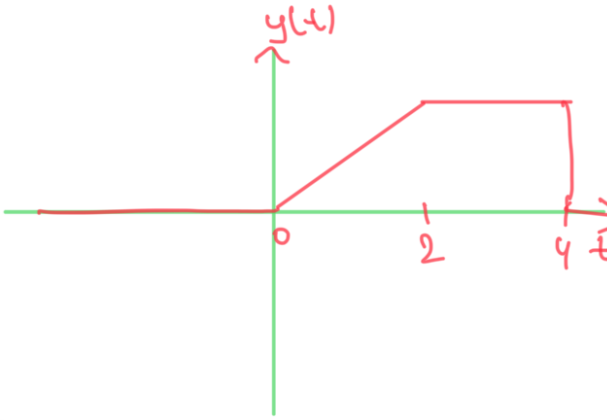
$$x(t) = \begin{cases} 1+t & ; -1 \le t \le 0 \\ 1 & ; 0 \le t \le 1 \\ 0 & ; \text{else} \end{cases} \qquad y(t) = x(t/2 - 1)$$





$$y(2t+2) = x(t)$$

---

even: $x(t) = x(-t)$ $\quad x[n] = x[-n]$

odd: $x(t) = -x(-t)$ $\quad x[n] = -x[-n]$

$$x_{even}(t) = \frac{x(t) + x(-t)}{2} \qquad x_{odd}(t) = \frac{x(t) - x(-t)}{2}$$

orthogonal: $\int_{-\infty}^{\infty} x(t)\, y^*(t) = \int_{-\infty}^{\infty} x^*(t)\, y(t) = 0$

$$\int_{-\infty}^{\infty} \left(\frac{e^{j\omega t} + e^{-j\omega t}}{2}\right)\left(\frac{e^{j\theta t} + e^{-j\theta t}}{2}\right) = \int_{-\infty}^{\infty} \frac{e^{j(\omega+\theta)t} + e^{-j(\omega+\theta)t}}{4}$$

$$+ \int_{-\infty}^{\infty} \frac{e^{j(\omega-\theta)t} + e^{-j(\omega-\theta)t}}{4}$$

$x(t) = x^*(-t)$ conjugate symmetry

$x(t) = -x^*(-t)$ conjugate asymmetry

$x(t) = Q(t) + jI(t)$

$x(t) = a(t) + jb(t)$

C.S $\Rightarrow$ $a(t) \rightarrow$ even $b(t) \rightarrow$ odd

C.A $\Rightarrow$ $a(t) \rightarrow$ odd $b(t) \rightarrow$ even

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$\rightarrow E_x + E_y = E_{x+y} \iff x, y$ are orthogonal

$\rightarrow P_x = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \qquad P_x = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{-N}^{N} |x[n]|^2$

$\rightarrow E_x < \infty \Rightarrow P_x = 0$

$\rightarrow P_x > 0 \Rightarrow E_x = \infty$

$\rightarrow P_x < \infty \Rightarrow$ Power signals

$\rightarrow$ causality: Output at time t depends only on the input at time t or earlier

$\rightarrow$ memoryless: output at time t depends only on the input at time t.

$\rightarrow$ Invertability: given an output signal, corresponding input can also be found; lossless transformations.

$\rightarrow$ Stability: Bounded input produces bounded output

$y(t) = (t^2 + 1)\, x(t) \rightarrow$ invertible, not stable

$y(t) = x^2(t) \rightarrow$ not invertible, stable

→ Linearity: a system that satisfies superposition.

$$f(ax_1(t) + bx_2(t)) = af(x_1(t)) + bf(x_2(t))$$

→ Time invariance: $f\{x(t)\} = y(t) \Rightarrow f\{x(t+T)\} = y(t+T)$

→ $x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(\tau - t)\,d\tau$

→ $y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)$

→ impulse response

$h(t)$ is output when input is $\delta(t)$

→ $y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$

→ $y(t) = x(t) * h(t)$ $\quad y[n] = x[n] * h[n]$

→ Steps:

(i) reverse $h(t)$

(ii) for each t find $h(t-\tau)$

(iii) multiply and add: $x(\tau)h(t-\tau)$

→ $x(t)$: $[-T_1, T_2]$ $\quad h(t)$: $[-T_2, T_4] \Rightarrow y(t)$: $[-T_1 - T_3, T_2 + T_4]$

→ convolution is commutative and associative

→ $x(t) * \delta(t - T) = x(t - T)$ ☆

→ $x(t) * h(t) = y(t) \Rightarrow x(t) * h(t - T) = y(t - T)$

> memoryless of LTI: $h(t) = A\delta(t)$

→ causality of LTI: $h(t) = 0$ for $t < 0$

→ invertibility of LTI: $h(t) * h^{-1}(t) = \delta(t)$

→ Stability of LTI: $\int_{-\infty}^{\infty} |h(t)|\,dt < \infty$

→ unit step response: $\int_{-\infty}^{t} h(t)\,dt$ (output for $x(t) = u(t)$)

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \qquad \omega_0 = 2\pi/T$$

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t)\, e^{-jk\omega_0 t}$$

↓

Spectrum of the signal → gives mag and phase of each phasor of $x(t)$ → gives frequency components of $x(t)$

→ Dirichlet conditions:

- → $x(t)$ must be absolutely integrable in a period
- → $x(t)$ must have finite no of maximas and minimas
- → $e(t)$ must have finite no. of discontinuities

→

Time domain	Frequency domain
even	even
odd	odd
C.S	real
C.A	img
real	C.S
img	C.A
$x(-t)$	a_{-k}
$x(t-s)$	$a_k e^{jsk\omega_0}$

$x^*(t)$ | a^*_{-k}

→ $E = T \sum_k |a_k|^2$ (in 1 time period)

→ $x(t) \cdot y(t) \rightarrow a_k^* b_k$

→ $x(t)^* y(t) \rightarrow a_k \cdot b_k \cdot T$

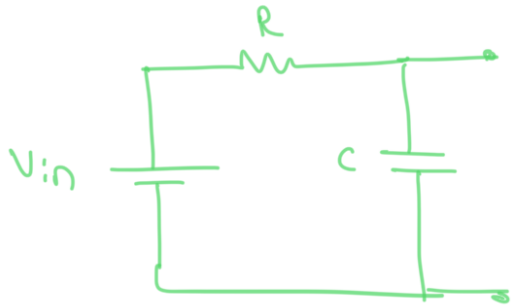
→ $H(j\omega) = \int h(\tau) e^{-j\omega\tau} d\tau$ (frequency response)

$x(t) * h(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$

→ differentiator: $d/dt\, x(t) \rightarrow a_k jk\omega \Rightarrow$ HPF

→ integrator: $\int_{-\infty}^{t} x(t)dt \rightarrow a_k / jk\omega \Rightarrow$ LPF

→



$Z_R = R \Rightarrow Z = R + 1/j\omega C$

$Z_C = 1/j\omega C$

$V_{out} = V_{in} \times \frac{Z_C}{Z_R + Z_C}$

$\Rightarrow H(j\omega) = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC}$

→ $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

→ $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$

→ $E_x = \int |x(t)|^2 = \frac{1}{2\pi} \int |X(j\omega)|^2 d\omega = \frac{E_x}{2\pi}$

$$x(t) = \text{rect}(t/T_0) = \begin{cases} 1 & t \in [-T_0/2, T_0/2] \\ 0 & \text{elsewhere} \end{cases}$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} = \int_{-T_0/2}^{T_0/2} e^{-j\omega t} = \frac{-1}{j\omega}\left(e^{-j\omega T_0/2} - e^{j\omega T_0/2}\right)$$

$$= \frac{2j\sin(\omega T_0/2)}{j\omega}$$

→ finite timelength ⇒ infinite bandwidth

→ infinite timelength ⇒ finite bandwidth

Time domain	Frequency domain	
$x^*(t)$	$X^*(-j\omega)$	no bandwidth change
$x(-t)$	$X(-j\omega)$	
$x(t-s)$	$X(j\omega)e^{-j\omega s}$	
$x(t)e^{j\theta t}$	$X(j(\omega-\theta))$	
$x(ct)$	$X(j\omega/c)/\lvert c\rvert$	$c>1 \Rightarrow$ expands, $c<1 \Rightarrow$ shrinks
even	even	
odd	odd	
real	C.S	
img	C.A	
C.S	real	
C.A	img	
$x(t).y(t)$	$X(j\omega) * Y(j\omega)/2\pi$	
$x(t)^*$		

$x(t)\ y(t)$ | $X(j\omega)\ Y(j\omega)$

Duality:

$$x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}X(j\omega)e^{j\omega t}d\omega$$

$$\Rightarrow 2\pi x(t)=\int_{-\infty}^{\infty}X(j\omega)e^{j\omega t}d\omega$$

$$\Rightarrow 2\pi x(-t)=\int_{-\infty}^{\infty}X(j\omega)e^{-j\omega t}d\omega$$

replace t with ω and ω with t

$$\Rightarrow 2\pi x(-\omega)=\int_{-\infty}^{\infty}X(jt)e^{-j\omega t}dt$$

$\Rightarrow$ FT of $X(jt)$ is $2\pi x(-\omega)$

IFT of $x(j\omega)=\frac{1}{2\pi}x(-t)$

$H(j\omega)=Y(j\omega)/X(j\omega)$

Bandwidth$\{y(t)\} \le \min\begin{cases}\text{Bandwidth of } x(t),\\ \text{Bandwidth of } h(t)\end{cases}$

→ Invertability: Inverse of $h(t)$ is IFT$\{1/H(j\omega)\}$

If $H(j\omega)=0$ for some $\omega \Rightarrow$ non invertable

→ Causality: any bandlimited $H(j\omega)$ is non causal

→ Stability: Dirichlet conditions (sufficient by not necessary)

→ $F\{\delta(t)\}=1$

$rect(t/\tau) \leftrightarrow \tau\,sinc\left(\frac{\omega\tau}{2\pi}\right)$

$x(t) \to X(\omega) \Rightarrow x(t) \leftrightarrow 2\pi x(-\omega)$

$\Rightarrow F\{1\} \to 2\pi\delta(\omega)$

$\Rightarrow F(C) \to 2\pi C\delta(\omega)$

→ IDEAL LPF:

$$H(f) = \begin{cases} 1 & |f| \le f_c \\ 0 & |f| > f_c \end{cases} \qquad h(t) = \frac{1}{2\pi}\int_{-f_c}^{f_c} e^{j\omega t}\,d\omega = \frac{1}{2\pi jt}\cdot 2j\sin(f_c t)$$

$$= \frac{1}{\pi t}\sin(f_c t)$$

non causal ; not stable ; not invertible

→ $x(t) = \text{sgn}(t)$

$$X(j\omega) = \int_{-\infty}^{0} (-1)e^{-j\omega t}\,dt + \int_{0}^{\infty} (1)e^{-j\omega t}\,dt$$

$$= -\int_{0}^{\infty} e^{j\omega t}\,dt + \int_{0}^{\infty} e^{-j\omega t}\,dt = -\int_{0}^{\infty} e^{-(a-j\omega)t}\,dt + \int_{0}^{\infty} e^{-(a+j\omega)t}\,dt$$

$$= 1/(a+j\omega) - 1/(a-j\omega) = 2/j\omega \quad (\text{let } a \to 0)$$

→ $u(t) = 1/2[1 + \text{sgn}(t)]$

⇒ $F\{u(t)\} = F\{1/2\} + 1/2\,F\{\text{sgn}(t)\} = \pi\delta(\omega) + 1/j\omega$

→ $F\{d/dt(x(t))\} = j\omega X(j\omega)$

→ $F\{\int_{-\infty}^{t} x(t)\,dt\} = X(j\omega)/j\omega + \pi X(0)\delta(\omega)$

$$\sum_{n=0}^{N} b_n \frac{d^n y(t)}{dt^n} = \sum_{m=0}^{M} c_m \frac{d^m x(t)}{dt^m} \Rightarrow Y(j\omega)\sum_{n=0}^{N} b_n (j\omega)^n = X(j\omega)\sum_{m=0}^{M} c_m (j\omega)^m$$

$$\Rightarrow H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{m=0}^{M} c_m (j\omega)^m}{\prod_{n=1}^{N} (d_n + j\omega)} = \sum_{n=1}^{N} \frac{f_n}{d_n + j\omega}$$

$$\Rightarrow h(t) = \sum_{n=1}^{N} f_n e^{-d_n t} u(t)$$

IFT of $\{a+j\omega\}^n = \frac{t^{n-1} e^{-at} u(t)}{(n-1)!}$

$$x(t) = y(t) + LC\frac{d^2y(t)}{dt^2} + RC\frac{dy(t)}{dt}$$

$$\Rightarrow X(j\omega) = Y(j\omega)\left[1 + LC(j\omega)^2 + RC(j\omega)\right]$$

$$\Rightarrow H(j\omega) = \frac{1}{1 + (RC)j\omega + LC(j\omega)^2}$$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$x(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} X(s) e^{st} ds$$

→ If ROC includes $s=0 \Rightarrow$ FT also exists

→ $x(t)$ is finite duration & bounded → ROC is entire s-plane

→ $x(t)$ is right sided and $\sigma = a$ is in ROC ⇒
↳ $\sigma > a$ is also in ROC

→ $x(t)$ is left sided and $\sigma = a$ is in ROC ⇒

$\sigma < a$ is also in ROC

→ $e^{-at}u(t) \rightarrow 1/s+a$ $Re\{s\} > -Re\{a\}$

$-e^{-at}u(-t) \rightarrow 1/s+a$ $Re\{s\} < -Re\{a\}$

→ $x(s) = N(s)/D(s)$

pole : roots of $D(s)$

zero : roots of $N(s)$

No of poles / zeros = degree of $N(s)$, $D(s)$

order = no of times that root repeats

→ $x(s) \rightarrow$ rational ⇒ no poles inside ROC

$x(t) \rightarrow$ right sided ⇒ ROC = right of rightmost pole

$x(t) \rightarrow$ left sided ⇒ ROC = left of leftmost pole

$x(t) \rightarrow$ 2 sided ⇒ ROC = strip b/w the poles